

Performance of DECT-2020 NR in an industrial environment for varying RF bands

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Abstract—As there are no publications regarding the performance of DECT-2020 NR in real-life scenarios yet, this work constitutes a first performance assessment of DECT-2020 NR in the 450, 1350 and 3700 MHz frequency bands at different numerologies with bandwidths ranging from 1.728 to 13.824 MHz. To show the benefits of DECT-2020 NR, it is compared to classic DECT in a Rayleigh fading channel. In the course of this evaluation, performance metrics such as Bit Error Rate (BER) and Packet Error Rate (PER) were considered. Furthermore, MATLAB-generated DECT-2020 NR packets were transmitted utilizing GNU Radio and Ettus USRPs. The measurements were conducted in an industrial production line. Overall, the packet error rates were found to be the best at 450 MHz, with a packet loss of less than 5 %. At 1350 MHz, a packet error rate of 15 % was observed, whereas at 3700 MHz, a maximum value of 10 % was recorded. The differences in PER for the different frequencies are attributed to the internal losses at higher frequencies in the used USRP B210s. Another aspect of this work focuses on characterizing the indoor channel of the production line using OFDM-based channel sounding. On average, an RMS delay spread of 65 ns was measured for a bandwidth of 13.824 MHz in the selected frequency bands.

Index Terms—DECT-2020, 5G, DECT, channel sounding, power delay profile, Doppler spectral density, channel statistics

I. INTRODUCTION

DECT-2020 NR, a new Radio Interface Technology (RIT) published and maintained by the European Telecommunications Standards Institute (ETSI), was evaluated and approved to be an IMT-2020 (5G) technology in 2021 [1]. The standard focuses on the IMT-2020 use cases of massive Machine Type Communications (mMTC) and Ultra-Reliable and Low Latency Communications (URLLC). Therefore, the technology targets the Industrial Internet of Things (IIoT) as well as application scenarios from the Program Making and Special Events (PMSE) sector.

The standard describes a non-cellular radio technology which allows independent, nomadic and self-organizing network deployments. Following the decentralized approach of mobile ad hoc networks, a Radio Device (RD) acts as a Fixed Termination point (FT) and/or as a Portable Termination point (PT). The FTs serve as access points and the PTs are the devices which connect and associate to the FTs. The Technical Specification (TS) is divided into four layers, namely the physical (PHY) layer, the medium access (MAC) layer, the data

link control (DLC) and convergence (CVG) layer [2]. On the physical layer, DECT-2020 NR utilizes channel coding for error correction and supports a vast range of bandwidths by adjusting the transmission numerology consisting of the subcarrier and Fourier transform scaling factors μ and β , respectively. The technology can achieve bandwidths between 1.782 and 221.184 MHz. Moreover, the standard allows a license-exempt operation in the DECT frequency band (1880 to 1900 MHz in the EU) which at the moment is exclusively available to classic DECT devices (e.g., telephones). Constituting an IMT-2020 technology, operation in the IMT frequency bands below 6 GHz is also possible.

In this paper, the performance of DECT-2020 NR is evaluated in frequencies which are approved in Germany for the PMSE sector (470 and 1350 MHz) and for the campus networks (3700 MHz) by utilizing different numerologies. Since a maximum bandwidth of 20 MHz is specified for wideband audio systems [3], the bandwidth is limited to 13.824 MHz in the measurement setup, as this is the closest possible bandwidth in DECT-2020 NR. The measurements were conducted in an industrial environment by using two USRP B210 Software-Defined Radios (SDR). Moving the receiver during the measurements introduces a time-varying channel. Additionally, the wireless doubly-selective channel is measured by using the setup as a channel sounder.

II. DECT-2020 NR VS CLASSIC DECT

The results were obtained by implementing the physical layer of the classic Digital Enhanced Cordless Telecommunications (DECT) standard by means of a link-level simulation. This link-level simulation achieves the same results for configuration 1a which was the focus of the evaluation in [4]. The DECT-2020 NR results were achieved by using an existing link-level simulator [1], [5]. Fig. 1 and Fig. 2 show the simulated packet error rates for a Rayleigh channel model (ITU-i in [6]) with a delay spread of 100 ns for classic DECT and DECT-2020 NR packets. The simulations apply to a single DECT channel (i.e., a bandwidth of 1.728 MHz) and packets occupying a single slot which has a duration of 416.67 μ s. Both simulations were over-sampled to 27.648 MHz and conducted under the assumption of a perfect time and frequency synchronisation. The dashed curves refer to the packet error rate of classic DECT that was

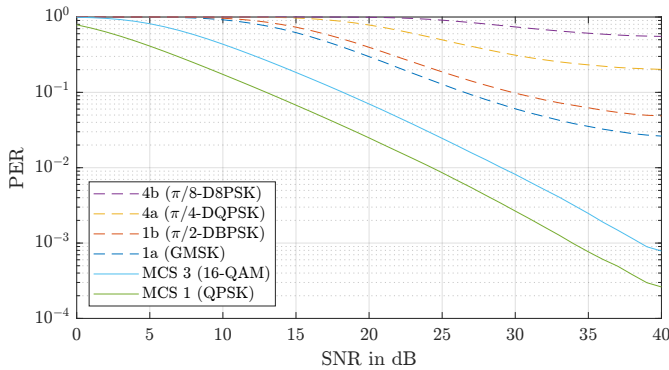


Fig. 1: Simulated packet error rates in a Rayleigh fading channel with one receiving antenna.

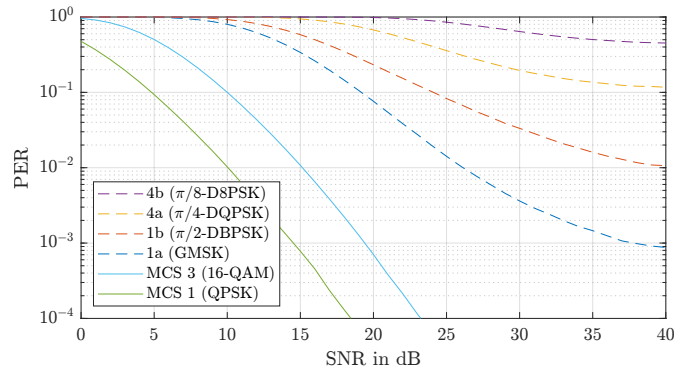


Fig. 2: Simulated packet error rates in a Rayleigh fading channel with two receiving antennas.

calculated on the 16-bit checksum of the control field (A-Field) which is more sensitive to errors than the 4-bit checksum in the data field (B-Field). Four different configurations of the classic DECT standard from the physical layer specification [7] are shown in Fig. 1 and Fig. 2, which determine the used modulation type. In this case only configuration 1a, which refers to Gaussian Minimum Shift Keying (GMSK) as modulation scheme, uses error correction in the form of a Viterbi decoder. For the other classic DECT configurations 1b, 4a and 4b, which refer to the modulation types $\pi/2$ -Differential Binary Phase Shift Keying ($\pi/2$ -DBPSK), $\pi/4$ -Differential Quadrature Phase Shift Keying ($\pi/4$ -DQPSK) and $\pi/8$ -Differential 8-Phase Shift Keying ($\pi/8$ -D8PSK) respectively, no channel coding was used. The two solid color curves refer to the DECT-2020 NR simulation where the packet error rate is calculated on the Physical Data Channel (PDC) since the Physical Control Channel (PCC) only uses Binary Phase Shift Keying (BPSK). The DECT-2020 NR simulation uses a Wiener filter for equalization and a turbo coder for error correction. In this case, MCS 1 uses Quadrature Phase Shift Keying (QPSK) at a coding rate of 1/2, which results in a Transport Block Size (TBS) of 296 bits. MCS 3 refers to 16-Quadrature Amplitude Modulation (16-QAM) and a coding rate of 1/2, which results in a TBS of 616 bits. The TBS corresponds to the number of user bits on the physical layer. These packet formats were chosen because they have a size comparable to the 320 (for GMSK and $\pi/2$ -DBPSK) and 640 (for $\pi/4$ -DQPSK) user bits in a classic DECT packet.

Fig. 1 shows the simulation results for one receiving antenna. For classic DECT, the GMSK configuration shows the best performance since it uses a Viterbi decoder for error correction. However, it reaches an error floor at approx. 10^{-3} . The other classic DECT formats also reach such an error floor with an overall poorer performance due to a higher number of bits per symbol. Overall, the DECT-2020 simulation shows a massive improvement in comparison to classic DECT, even when using a higher modulation scheme (MCS 3, i.e., 16-QAM at a coding rate of 1/2). Additionally, in the range of 0 to 40 dB, the error rates do not reach an error floor because OFDM is more robust to fading. When comparing packets in the same order of user

bits (configuration 1a and MCS 1), it can be seen that the MCS 1 curve is shifted by approximately 15 dB to the left.

The results change drastically when using receiver diversity, which is illustrated in Fig. 2. In this case, Maximum Ratio Combining (MRC) is used for both simulations. However, the DECT-2020 NR algorithm is more efficient because it uses the channel equalization for weighting the antenna streams. In this case, the DECT-2020 NR curves move to the left by about 15 dB as compared to when using no receiver diversity. In the case of classic DECT, there is also a performance gain which is only noticeable for the configurations 1a and 1b. For the higher order modulation schemes the performance gain is almost negligible.

III. WIRELESS CHANNEL

A. Channel Modelling

The wireless channel describes how a transmitted signal gets distorted by path loss, shadowing and fading. Fading is caused by multipath components which are added up at the receiver. The cause for multipath components is scattering by objects. This is described in the following Eq. 1, where $y(t)$ is the received signal, $x(t)$ is the transmitted signal, $h(t)$ the channel impulse response and $n(t)$ the thermal noise which is added at the antenna of the receiver. The impact of the wireless channel is depicted by $h(t)$.

$$y(t) = h(t) * x(t) + n(t) \quad (1)$$

The channel can be regarded as a linear system that follows a Wide-Sense Stationary Uncorrelated Scattering (WSSUS) model [8], [9]. The wide-sense stationary part refers to the statistical properties of the channel which are in this case considered to change very slowly. Therefore, the channel is seen as wide-sense stationary over short periods. The uncorrelated scatterer part means that the reflected signals from the scatterers are not correlated. The channel impulse response itself refers to a declining exponential function which is usually represented as a tapped-delay line model in Eq. 2.

$$h(\tau, t) = \sum_{n=1}^N \beta_n(t) e^{j\theta_n(t)} \delta(t - \tau_n(t)) \quad (2)$$

Each of the N scattered waves reaches the receiver with an amplitude $\beta_n(t)$, a phase $\theta_n(t)$ and a delay of $\tau_n(t)$. Applying the Fourier transform on the dimension of τ to the channel impulse response $h(\tau, t)$ results in the channel transfer function $H(f, t)$.

$$H(f, t) = \mathcal{F}\{h(\tau, t)\} \quad (3)$$

The channel statistics are derived from the correlation function $R_H(t, t + \Delta t, f, f + \Delta f)$ of the channel transfer function which is described in the following Eq. 4 [10] where $E\{\cdot\}$ represents the expected value.

$$R_H(t, t + \Delta t, f, f + \Delta f) = E\{H^*(t, f)H(t + \Delta t, f + \Delta f)\} \quad (4)$$

Since the channel is considered WSSUS, the correlation function does not depend on the absolute time t and the frequency f which is shown in the following term.

$$R_H(t, t + \Delta t, f, f + \Delta f) = R_H(\Delta t, \Delta f) \quad (5)$$

The power delay profile $P_h(\tau)$ is the function from which the RMS delay spread S_τ is derived. There are two ways to obtain the power delay profile. The first option is to average the channel impulse response over time t , which can be done because the channel is considered WSSUS. The second option is to use the inverse Fourier transform on the frequency dimension of the channel correlation function $R_H(\Delta t, \Delta f)$ as shown in Eq. 6.

$$P_h(\tau) = \mathcal{F}^{-1}\{R_H(\Delta t, \Delta f)|_{\Delta t=0}\} \quad (6)$$

The delay spread S_τ is defined as the second central moment of the power delay profile $P_h(\tau)$. The Eq. in 7 is for discrete values.

$$S_\tau = \sqrt{\frac{\sum_{i=1}^N (\tau_i - T_D)^2 P_h(\tau_i)}{\sum_{i=1}^N P_h(\tau_i)}} \quad (7)$$

$$\text{with } T_D = \frac{\sum_{i=1}^N P_h(\tau_i) \tau_i}{\sum_{i=1}^N P_h(\tau_i)} \quad (8)$$

The RMS delay spread S_τ should also not be calculated over all signal components of the power delay profile but instead only over a limited delay interval [11]. The delay interval is defined as the fraction of the power delay profile up to the last significant component and can be set as the period in which the power delay profile exceeds a certain threshold value at the first time and falls below this threshold at the last time. A typical threshold value is, e.g., 20 dB below the maximum of the power delay profile $P_h(\tau)$.

The RMS Doppler spread S_ν is calculated from the Doppler power spectral density $P_B(\nu)$ which is derived from the channel correlation function by using the Fourier transform on the time Δt dimension where ν represents the Doppler frequencies.

$$P_B(\nu) = \mathcal{F}\{R_H(\Delta t, \Delta f)|_{\Delta f=0}\} \quad (9)$$

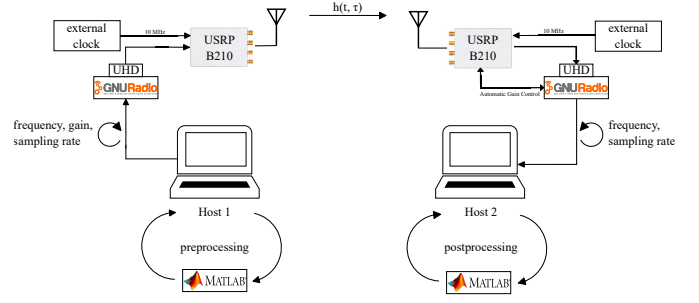


Fig. 3: Measurement setup

TABLE I: Measurement parameters

Attribute	Value
Signal bandwidth [MHz]	1.728, 3.456, 6.912, 13.824
Carrier frequency [MHz] (antenna reflection coefficient [dB])	450 (-11.5), 1350 (-9.5), 3700 (-21)
MCS	1 (QPSK)
Receiver equalizer	Wiener filter
CFO and STO correction	active
Measurement duration	15 s

The RMS Doppler spread is the second central moment which is calculated in Eq. 10 for discrete values.

$$S_\nu = \sqrt{\frac{\sum_{i=1}^N (\nu_i - T_{B,m})^2 P_B(\nu_i)}{\sum_{i=1}^N P_B(\nu_i)}} \quad (10)$$

$$\text{with } T_{B,m} = \frac{\sum_{i=1}^N P_B(\nu_i) \nu_i}{\sum_{i=1}^N P_B(\nu_i)} \quad (11)$$

B. Channel Sounding with OFDM

Channel sounding can be done by using an OFDM signal which is showcased in [12]. The channel is then obtained by using the received signal in the frequency domain after the cyclic prefix is removed and equalizing it with the original transmitted signal in the frequency domain. This is represented in the following Eq. 12, where $Y(t, f)$ and $X(t, f)$ denote the received and transmitted signals in the frequency domain, respectively.

$$H(t, f) = Y(t, f)/X(t, f) \quad (12)$$

IV. MEASUREMENT CONFIGURATION

A. Measurement Setup

The measurement setup is shown in Fig. 3. For every DECT-2020 NR numerology, packets are pre-generated as binary files via the link-level simulation [5]. Depending on the numerology, the packets are between eight and 24 slots in length. The files are repeatedly transmitted (without pauses) via an USRP B210 which is controlled by GNU Radio. The receiver side also utilizes a USRP B210 with GNU Radio where the samples are stored in a binary file. During a measurement, the receiver gain is continuously controlled by an automatic gain control (AGC), which is implemented as a GNU Radio block [13]. The AGC prevents clipping by ensuring that the ADCs of the USRPs

TABLE II: Transmitter output power in dBm

μ	β	Bandwidth [MHz]	450 MHz [78 dB]	1350 MHz [78 dB]	3700 MHz [80 dB]
1	1	1.728	7	4	-6
1	2	3.456	7	4	-6
1	4	6.912	4	1.5	-8.5
1	8	13.824	1.5	-1	-13
2	1	3.456	6	3.5	-6
2	2	6.912	5.5	3	-7
2	4	13.824	2	-1	-5
4	1	6.912	4	1.5	-10
4	2	13.824	1.5	-1	-10
8	1	13.824	2	-0.5	-10.5

TABLE III: Calculated channel statistics

Frequency [MHz]	450	1350	3700
RMS Doppler spread [Hz]	1.68	4.27	13.32
RMS delay spread [ns] (avg.)	65.34	67.86	64.36
RMS delay spread [ns] (min.)	7.18	7.22	7.26
RMS delay spread [ns] (max.)	150	160	185

are working on a balanced level. The receiver gain values are also written to a file while a measurement is running and later used for calculating the channel. The stored binary file is then processed offline in MATLAB. Both USRPs are using an external rubidium oscillator as a reference clock because the accuracy of the internal clock (2 ppm [14]) is not sufficient. The measurement parameters are depicted in Table I.

The transmitter signal power varies depending on the carrier frequency due to internal losses inside the USRP (output power at 6 GHz is almost 20 dB lower than the output power at 600 MHz). Furthermore, the highest transmitter gain level is shown to result in a distorted output signal (this is described in detail in [15]). For the measurement, the gain is set to a level at which the output signal is mostly free of this distortion. Moreover, the signal power has to be decreased when increasing the signal bandwidth (i.e., increasing the number of subcarriers, β in DECT-2020 NR) due to the higher Peak-to-Average Power Ratio (PAPR). The measured output power in dBm for the used numerologies is shown in Table II.

B. Measurement Scenario

The described measurement setup was used in a production line (depicted in Fig. 4) at the Institute of Production Engineering and Machine Tools (IFW) which is part of the mechanical engineering campus at Leibniz University Hannover. The measurements were conducted by moving the receiver down the production line in walking speed which takes approximately 15 seconds. The transmitter is kept stationary in the middle of the line.

V. RESULTS

A. Performance Measurements

The results of the measurements are shown in Fig. 5. The Bit Error Rates (BERs) below 0.1 % and Packet Error Rates (PERs) below 5 % in Fig. 5(a) and 5(d) in the 450 MHz spectrum are the lowest. This was expected because of the

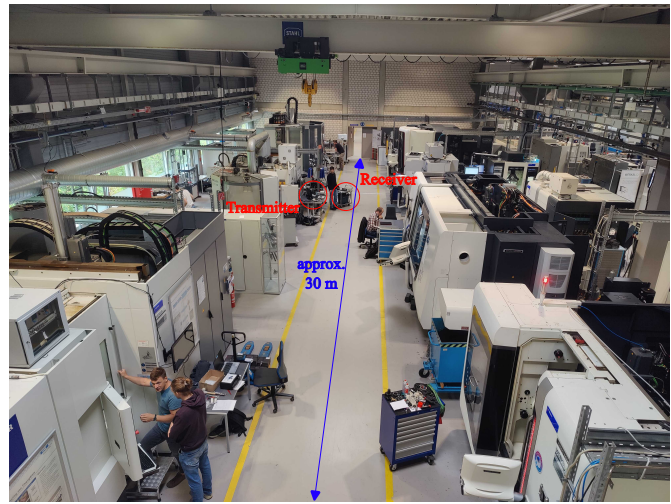


Fig. 4: The production line at the IFW.

higher output signal power at the lower carrier frequencies and therefore a overall better Signal-to-Noise Ratio (SNR) at the receiver. The error rates for the measurements in 1350 and 3700 MHz are higher (maximum BER of 0.4 % and maximum PER of 13 % in the 1350 MHz frequency band) due to a lower SNR. When comparing the error rates in the 1350 and 3700 MHz bands, it is noticed that the error rates are almost on the same level (which is especially true for the bit error rates in Fig. 5(b) and 5(c)). Due to the decreased output power level at 3700 MHz and therefore a lower SNR at the receiver, the error rates at this carrier frequency were expected to be the overall lowest.

When observing the BER or PER for a frequency band, it shows that the error rates increase when using a higher number of subcarriers (e.g., Fig. 5(a): $(\mu, \beta) = (1, 2), (1, 4)$, etc.). This is due to the channel which is more or less flat for the lower bandwidths and becomes frequency selective for higher bandwidths (especially for 13.824 MHz) which makes the channel estimation more difficult for the Wiener filter and results in a higher error rate. This can be verified in the derived channel transfer functions in Fig. 6. The channel transfer functions for 1.728 MHz show little to no changes on the frequency axis for a time point (e.g. Fig. 6(a)), while the channels with the highest bandwidth (e.g., Fig. 6(d)) show variety due to fading. Therefore, the 1.728 MHz signal bandwidth can be interpreted to be below the coherence bandwidth. It is also noticeable that numerologies of the same bandwidth differ significantly in their error rate (e.g. $\mu = 1, \beta = 8$ and $\mu = 8, \beta = 1$ in Fig. 5(b)). This is due to a shorter cyclic prefix when using wider subcarriers which makes the transmission more vulnerable to inter-symbol interference if the channel impulse function is longer than the cyclic prefix. In addition, using more subcarriers instead of an increased subcarrier bandwidth also raises the number of reference signals which provide more information to the Wiener filter resulting in a better channel estimation.

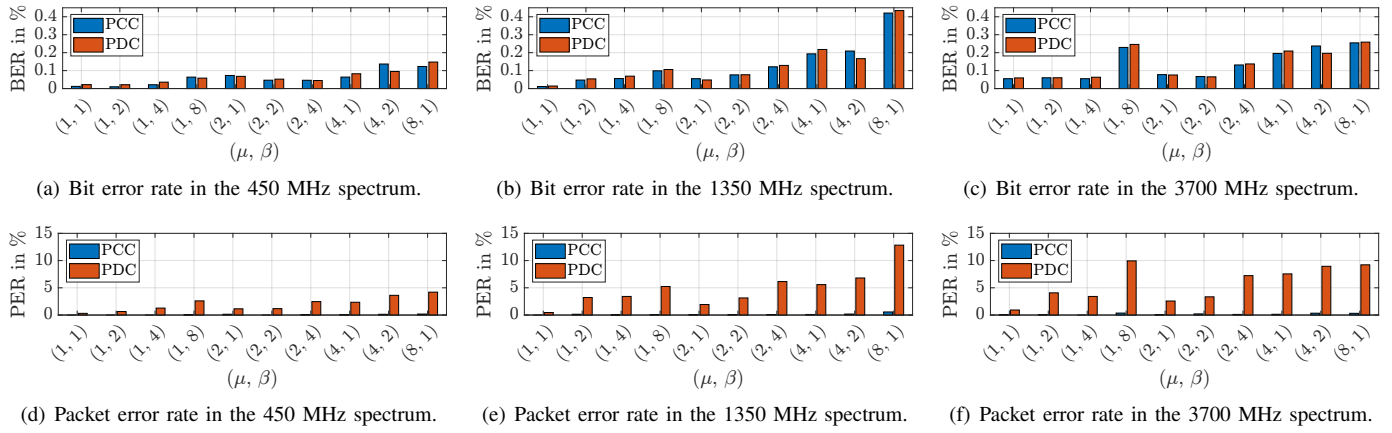


Fig. 5: Bit and packet error rates.

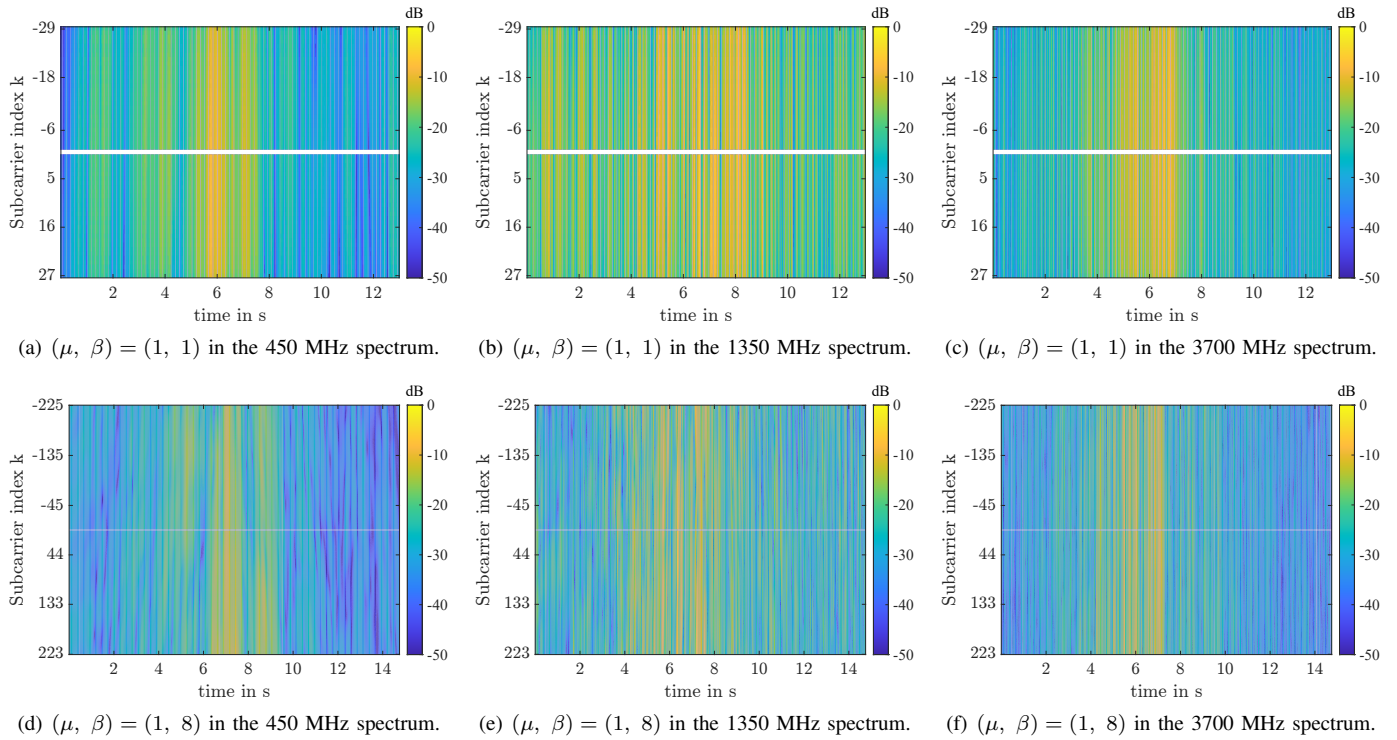


Fig. 6: Channel transfer functions for bandwidths of 1.728 MHz and 13.824 MHz (in the first and second row, respectively).

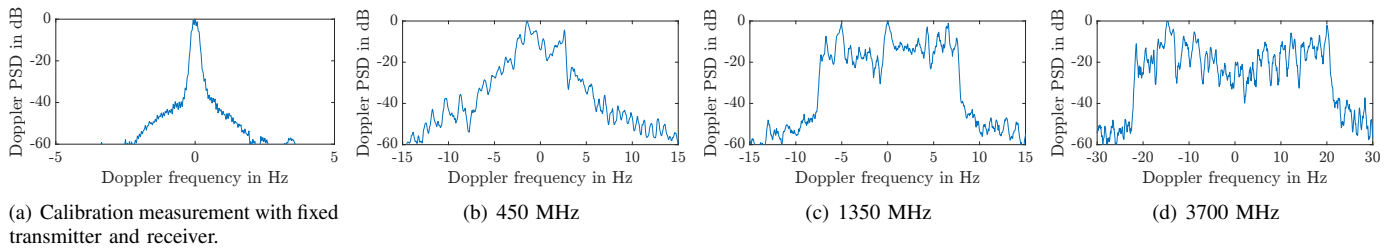


Fig. 7: Doppler power spectral density

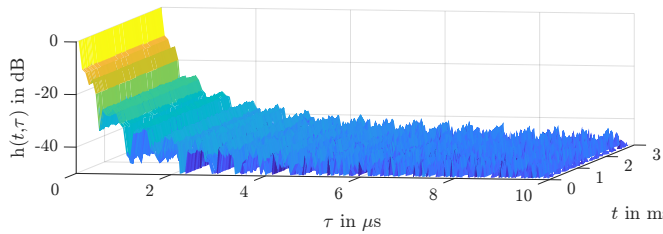


Fig. 8: Channel impulse response for $(\mu, \beta) = (1, 8)$ in the 450 MHz spectrum.

B. Channel Sounding

In Fig. 6, the measured channels for the 450, 1350 and 3700 MHz frequency bands are shown as surface plots. The vertical axis represents the subcarriers, each having a bandwidth of 27 kHz. The channel transfer functions are corrected by the measured receiver gains to highlight the fading effect. The maximum at 6-8 seconds represents the point in time where the receiver passes the transmitter. The channels show a different fading behavior which can be attributed to the wavelength of the carrier frequency. Also, as previously mentioned, the signal bandwidth of 1.728 MHz for the numerology $\mu = 1, \beta = 1$ shows very little variety on the frequency axis for a time point which allows the conclusion that this bandwidth is below the coherence bandwidth. The opposite can be seen for the signal bandwidth of 13.824 MHz in the Fig. 6(d), 6(e) and 6(f) which show frequency-selective behaviour.

The derived Doppler spectral density can be seen in Fig. 7 where Fig. 7(a) represents a calibration measurement with a stationary receiver. Since the receiver is moving in a direct line towards and away from the transmitter, the Doppler spectrum is expected to spread up to the maximum Doppler frequency. The spreading of the spectrum depends on the carrier frequency which determines the maximum Doppler frequency as can be clearly seen in Fig. 5(b) and 5(c). This can be confirmed by calculating the Doppler frequency for the receiver moving with a velocity of approximately $2 \frac{m}{s}$. If the receiver is not moving, the Doppler spectrum is expected to show no spreading which is verified in the calibration measurement in Fig. 7(a). The RMS Doppler spreads, which were calculated according to Eq. 10 for the used frequency bands, are shown in table III. For the calculation, only values of the spectral densities above a defined threshold (10 dB below the maximum) were used.

In Fig. 8 the channel impulse response in the 450 MHz frequency band is shown. It can be seen that the impulse response falls below the 20 dB threshold before $1 \mu s$. The channel impulse responses for 1350 and 3700 MHz look similar which is represented in the average RMS delay spread according to Eq. 8 in Table III which shows that the average RMS delay spread is the same for all three frequency bands. The calculation of the RMS delay spread is based on the numerology $\mu = 1, \beta = 8$ since it has the highest resolution for the delay τ . A difference for the measured delay spreads can be seen for the maximum delay spread of 185 ns in the 3700 MHz band.

VI. CONCLUSION

To make a first performance evaluation of DECT-2020 NR in different frequency bands, measurements were conducted in a real industrial setup. The measurements show packet error rates less than 5 % for 450 MHz, less than 15 % for 1350 MHz, and less than 10 % for 3700 MHz depending on the DECT-2020 numerology which determines the bandwidth of the signal. In addition, the channel statistics were derived by using the OFDM system as a channel sounder. The channel sounding of the industrial environment resulted in a measured RMS delay spread of approximately 65 ns on average for all frequency bands and a maximum delay spread of 180 ns for the 3700 MHz spectrum. In classic DECT, these channel conditions would cause severe packet losses.

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